

Slides 2.18: Buying Low, Rebalancing, and Return Math

Lesson 2.18, one day. Slides 1 to 12. Write “Round 8, 1932” and the prices on the board before class from page 2 of Handout 02.14 - The Gilded Age Market Kit.md and keep them covered until slide 3; the dispatch is on page 3.

Slide 1: Do now

- In 1928, GAS was \$40 a share. Suppose it fell to \$20 and you bought 10 shares for \$200.
- If the price ever climbs back to \$40, how much do you make?
- Harder: what percent of your money did you make? Notes: 3 minutes. \$20 a share, \$200 on 10 shares. Percent: 200 divided by 200, which is 100 percent. You doubled your money.

Slide 2: That is what buying low means

- It is also what everyone who sells at the bottom hands to the person who buys from them.
- Today: the one formula that tells you who won. Image: two figures in 1930s coats passing a stock certificate; the one handing it over looks scared, the one receiving it looks calm.

Slide 3: Round 8. 1932. The bottom.

- RAIL 10. OIL 25. GAS 20. ELEC 10. STEEL 25. AUTO 8. RADIO 3.
- Prices have fallen for almost three years. On July 8, 1932, the Dow Jones average closed at 41, eighty nine percent below its 1929 peak.
- About one worker in four has no job. Thousands of banks have failed and the money in them is gone. (Our bank survives. Most did not.)
- General Motors sells for what a good shirt costs. Image: a 1932 bread line under a shop sign. Notes: Uncover the prices. Ledgers first: the Round 8 column and the holdings table. Then read the dispatch from kit page 3 and ask its question: everyone is selling and nobody is buying. If you have cash, what does that make this? 5 minutes. Do not open trading yet.

Slide 4: The bottom, and buying low

- The bottom: the lowest price before a recovery (the climb back). Nobody knows it until it is behind them.
- Buying low: buying a company after it fell, because you believe it will recover.
- The risk: some do not. The three tests still apply. A cheap price is not a reason. A dispatch with a number is. Image: a line graph with a V and a question mark at the bottom of the V.

Slide 5: Rebalancing after a crash

- Your pie changed shape without you doing anything: the risky slices shrank.

- Rebalancing: buy a little of what fell, sell a little of what held, to get back to the pie you chose in Round 5.
- Regular upkeep, not a hero move. Image: a pie with two shrunken slices, an arrow, and the same pie restored. Notes: Students copy the three definitions to the journal's My notes page.

Slide 6: The formula

- Percent return = (ending minus starting) divided by starting, times 100.
- The bottom of the fraction is always the starting value.
- A loss gets a minus sign. Image: the fraction drawn large: "ending minus starting" over "starting," with "x 100" beside it and STARTING in bold.

Slide 7: Worked example

- Started with \$1,000 in 1870. \$850 now.
- 850 minus 1,000 = minus 150.
- Minus 150 divided by 1,000 = minus 0.15.
- Times 100: minus 15 percent. Notes: Say "divided by the starting value" three times today.

Slide 8: Return Math, Parts A and B

- Part A: four practice problems from the board's prices, alone. Then the trick question: GAS, 40 to 20 and back.
- Part B: my own return, from my ledger's Round 8 total, alone. And the company that hurt me most, 1928 price to 1932 price.
- Part C waits until after trading. Notes: 12 minutes for slides 4 to 8. The common error is dividing by the ending value. Watch for it at the desk.

Slide 9: Trading, Round 8, then Part C

- Trading is open for four minutes at 1932 prices.
- If your pie changed shape in the crash, this is the round to fix it.
- If you are buying low, write which company and what you expect on Entry 8 first. If you are holding, write that too.
- At four minutes the market freezes. Nobody trades again until the closing bell.
- Then Part C with a partner: three investors from 1870 who never traded, at the peak, in the crash, and today. Notes: 10 minutes. Four for trading, six for Part C.

Slide 10: Debrief Part C

- Safe, all in the bank: plus 19.5, plus 23.1, plus 26.8 percent. Never dropped.
- One company, all in RAIL: plus 40.0, minus 10.0, minus 80.0 percent.
- Mixed, four ways: plus 189.1, plus 117.9, plus 24.5 percent.
- Which would you want for college in ten years? For rent next month? Image: three lines on one graph across 1928, 1929 and 1932: a gentle rise, a cliff, a mountain that comes most of the way down. Notes: Say the honest thing: the bank won today by two

points after 62 years, and the mixed ledger was ahead in the other two rounds on the table (the peak and the crash).

Slide 11: Entry 8

- What I did (rebalance, buy low, or hold). Why. What happened.
- My percent return so far is ___ (from Part B).
- The company that hurt me most fell ___ percent from 1928 to 1932.
- My total in dollars at the freeze. Notes: 6 minutes.

Slide 12: Closure

- Exit card: the percent return formula in words. My return today is ___ percent. The all-bank ledger's return today is ___ percent. One sentence: which is better, and better for what?
- Three investors read the last sentence.
- Next class: the closing bell, 1936. And what happened next to every company you own. Notes: 4 minutes. The market is frozen at Round 8. Do not say what 1936 prices look like.